

Laboratory Exercise

Geometric Transformations

Objective:

At the end of the exercise, the students should be able to:

- Scale, translate, and rotate an object.

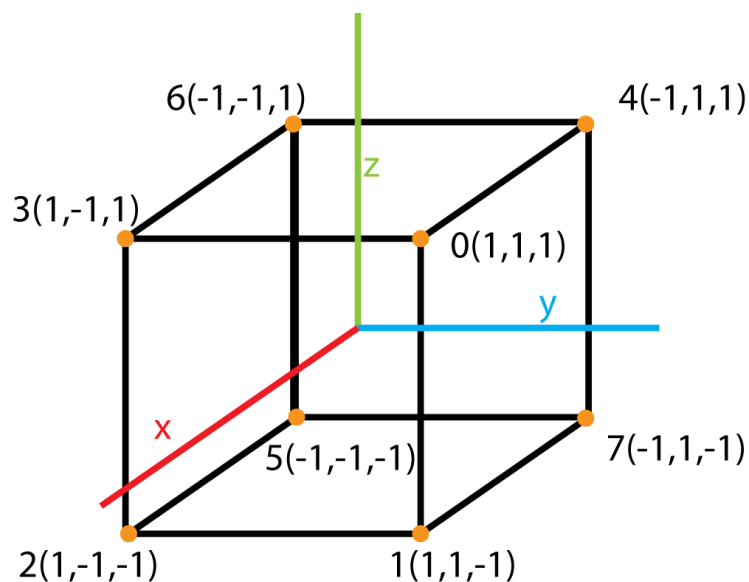
Software Requirement:

- Python 3.7 or higher

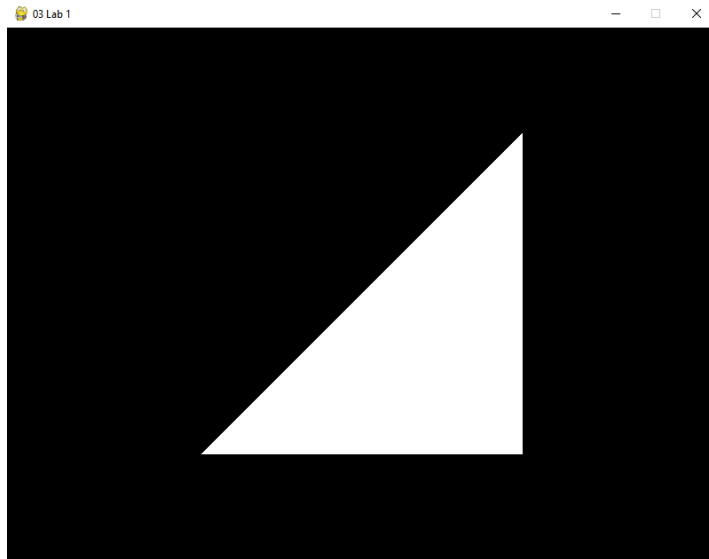
Procedure:

1. Launch **IDLE** and open your Python module from **03 Laboratory Exercise 1**.
2. Save a copy of your module using a different filename, then run it by clicking **Run > Run Module** or by pressing **F5**.
3. Change the window's caption to **"04 Lab 1"**.
4. Redraw your cube using 12 triangles. In your **draw_cube()** method, replace **GL_LINES** with **GL_TRIANGLES**, then edit the **glVertex3f()** statements based on the following table and figure.

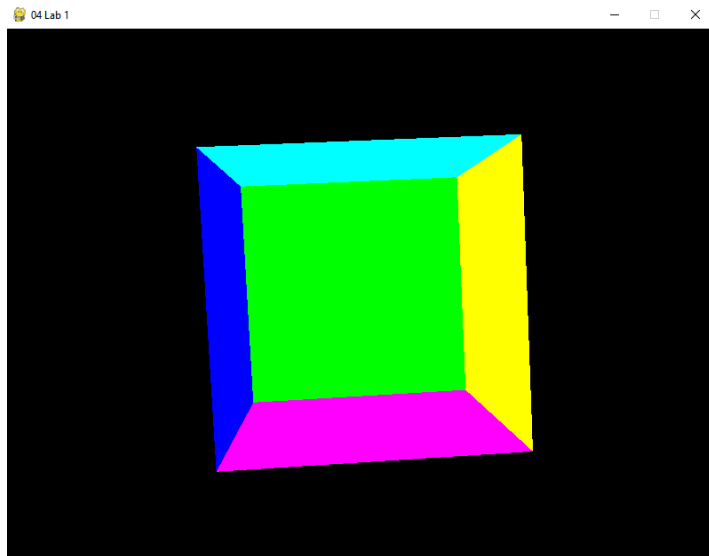
Vertex	X	Y	Z
0	1	1	1
1	1	1	-1
2	1	-1	-1
3	1	-1	1
4	-1	1	1
5	-1	-1	-1
6	-1	-1	1
7	-1	1	-1



For every set of three (3) vertices, you can form a triangle. For example, three (3) `glVertex3f()` statements can form a triangle, as illustrated in the sample output below.



5. Set a specific color for each pair of triangles using `glColor3f()`. The `glColor3f()` function has three (3) parameters for setting the RGB value. For example, `glColor3f(0,1,0)` draws in full-intensity green. Each `glColor3f()` statement is placed before every pair of triangles.



6. The `glScalef()` function multiplies the current matrix by a general scaling matrix. It has three (3) parameters for specifying the scale factors along the x, y, and z axes, respectively. Decrease the size of your cube using this function.
7. To enable the function of updating the depth buffer, insert this statement before the `gluPerspective` statement: `glEnable(GL_DEPTH_TEST)`
8. Apply geometric transformations to your cube using your keyboard. In the sample code below, the cube moves to the left if you press A.

```
if event.type == pygame.KEYDOWN:
    if event.key == pygame.K_a:
        glTranslatef(-1,0,0)
```

9. Show/submit your code and output to your instructor.
10. Save a copy of your code for future use.

Grading Rubric:

CRITERIA	PERFORMANCE INDICATORS	POINTS
Correctness	The geometric transformations applied to the cube are complete and correct.	30
Efficiency	The code is concise without sacrificing correctness and logic.	20
TOTAL		50